Pamela Ferretti

Postdoctoral Researcher
Department of Medicine
University of Chicago, Chicago, IL, USA

Google Scholar

EDUCATION

European Molecular Biology Laboratory (EMBL) — **Heidelberg, Germany**
Ph.D., Computational Biology — Dec 2021
Advisors: Peer Bork, Sebastian Schmidt

University of Trento — **Trento, Italy**
MSc, Cellular and Molecular Biotechnology — Mar 2016
Advisor: Nicola Segata

University of Trento — **Trento, Italy**
BSc, Computer Science — June 2014
Advisor: Nicola Segata

EMPLOYMENT

University of Chicago — **Chicago, IL, USA**
Postdoctoral Researcher — Feb 2023 - Present
Advisor: Ran Blekhman

European Molecular Biology Laboratory (EMBL) — **Heidelberg, Germany**
Bridging Postdoc — Jan 2022 – Jun 2022
Advisors: Peer Bork, Sebastian Schmidt

University of Trento — **Trento, Italy**
Research Assistant — Aug 2016 – Aug 2017
Advisor: Nicola Segata

RESEARCH INTERESTS

I use strain-level metagenomics and machine learning approaches to study:

- Human breast milk composition and its role in shaping infant microbiome assembly and development.
- Microbiome transmission dynamics within social groups in both human and animal populations.

PUBLICATIONS

Total number of publications: 27; Total citations: 4,510; H-Index: 17. (* indicates shared authorship)

Publications in peer-reviewed journals:

2026 Southworth C, Winans J, Gordon J, Learn N, Wilber W, Andreadis C, Andreasen G, Arandjelovic M, Campbell R, Chege M, Creighton M, Cromer C, Debray R, Dickson C, **Ferretti P**, George E, Gesquiere L, He S, Hey L, Jefferson E, Kulahci I, Lerch B, Nonnamaker L, Rivas-González I, Tuliozi B, Webb S, Alberts S, Archie E, and Tung J. *Demographic, behavioral, and ecological data from a long-term field study of wild baboons in Amboseli, Kenya.* Scientific Data

2025 **Ferretti P*,** Allert M*, Johnson K, Rossi M, Heisel T, Gonia S, Knights D, Fields D, Albert FW, Demerath EW, Gale CA, and Blekhman R. *Assembly, stability, and dynamics of the infant gut microbiome are linked to bacterial strains and functions in mother's milk.* Nature Communications*, in press*

2025 Kuhn M, Schmidt TSB, **Ferretti P**, Głazek A, Robbani M, Akanni W, Fullam A, Cetin E, Hassan, Noack K, Schwarz A, Thielemann R, Thomas L, Moritz von Stetten M, Alves R, Kartal E, Ivan Kel I, Iyappan

A, Maistrenko O, Mankowski A, Nishijima S, Keller MI, Podlesny D, Schulz S, Van Rossum T, and Bork P. *Metalog: manually curated metadata for global metagenomics samples*. Nucleic Acids Research, *in press*

2025 **Ferretti P**. *The gut remembers: the long-lasting effect of medication use on the gut microbiome*. mSystems

2025 **Ferretti P**, Martignoni MM, McManus L, Sakal T, Liaghat A, Stevens B, K, Souza L, Cardon Z, Silveira C, Bordenstein S, and Roughgarden J. *Theory of host-microbe symbioses: challenges and opportunities*. Cell Host & Microbe

2025 Chege M, **Ferretti P**, Webb S, Macharia RW, Obiero G, Kamau J, Alberts S, Tung J, Akinyi M, and Archie E. *Eukaryotic composition across seasons and social groups in the gut microbiota of wild baboons.* Animal Microbiome

2025 Hug AL, Hatzenichler R, Moraru C, Soares A, Meyer F, Heyder A, **The Data Reuse Consortium**, and Probst A. *A roadmap for fair reuse of public microbiome data.* Nature Microbiology

2025 **Ferretti P**, Johnson K, Priya S, and Blekhman R. *Genomics of host-microbiome interactions in humans.* Nature Reviews Genetics

2025 Abdill R*, Graham* S, Rubinetti V, Ahmadian M, Hicks P, McDonald D, Chetty A, **Ferretti P**, Gibbons E, Rossi M, Knight R, Krishan A, Frank , Albert FW, Casey S. Greene CS, Sean Davis S, and Ran Blekhman R. *Integration of 168,000 samples reveals global patterns of the human gut microbiome*. Cell

- Featured on the journal cover.

2024 Nishijima S, Stankevic E, Aasmets O, Schmidt TSB, Nagata N, Keller MI, **Ferretti P**, Juel H, Fullam A, Schudoma C, Robbani SM, Kragh Hansen J, Aas Holm L, Israelsen M, Schierwagen R, Broens C, Fonvig CE, Holm JC, Nielsen T, Steen Pedersen J, Thiele MS, Trebicka J, Org E, Krag A, Hansen T, Kuhn M, and Bork P. *Fecal microbial load is a major determinant of gut microbiome variation and a confounder for disease associations*. Cell

2023 Schmidt TSB*, Fullam A*, **Ferretti P**, Orakov A, Maistrenko OM, Ruscheweyh HJ, Duan Y, Van Rossum T, Sunagawa S, Mende D, Kuhn M, Coelho LP, and Bork P. *SPIRE: a Searchable, Planetary-scale, Integrated REsource of microbial diversity and function*. Nucleic Acids Research

2023 **Ferretti P**, Wirbel J, Maistrenko OM, Van Rossum T, Alves R, Fullam A, Akanni W, Schudoma C, Schwarz A, Thielemann R, Thomas L, Kuhn M, Zeller G, Schmidt TSB, and Bork P. *C. difficile may be overdiagnosed in adults and is a commensal in infants*. eLife

2021 Klopp J, **Ferretti P**, Meyer C, Hilbert K, Haiß A, Marißen J, Henneke P, Hudalla H, Pirr S, Viemann D, Zemlin M, Forslund SF, Härtel C, Bork P, Van Rossum T, and the PRIMAL consortium. *Meconium microbiome of very preterm infants across Germany*. mSphere

2021 Coelho LP, Alves R, del Río AR, Myers PN, Cantalapiedra CP, Giner-Lamia J, Schmidt TSB, Mende D, Orakov A, Letunic A, Hildebrand F, Van Rossum T, Forslund SF, Khedkar S, Maistrenko OM, Jia L, **Ferretti P**, Sunagawa S, Zhao ZM, Bjørn Nielsen H, Cepas JH, and Bork P. *Towards the biogeography of prokaryotic genes*. Nature

2021 Hildebrand F, Gossmann T, Frioux C, Özkurt E, Myers PN, **Ferretti P**, Bahram M, Bjørn Nielsen H, and Bork P. *Dispersal strategies shape persistence and evolution of gut bacteria.* Cell Host & Microbe

2021 Bahram M, Netherway T, Frioux C, **Ferretti P**, Coelho LP, Geisen S, Bork P, and Hildebrand F. *Global distribution and diversity of bacteria and fungi across diverse habitats.* Environmental Microbiology

2020 Van Rossum T*, **Ferretti P***, Maistrenko OM*, and Bork P. *Diversity within species: interpreting strains in microbiomes.* Nature Reviews Microbiology

- Featured on the journal cover

2018 **Ferretti P**, Pasolli E, Tett A, Asnicar F, Gorfer V, Fedi S, Armanini F, Truong DT, Manara S, Zolfo M,

Beghini F, Bertorelli R, De Sanctis V, Bariletti I, Canto R, Clementi R, Cologna M, Crifò T, Cusumano G, Gottardi S, Innamorati C, Masè C, Yassour M, Vlamakis H, Xavier R, Koren O, Tateo S, Soffiati M, Pedrotti A, Ventura M, Huttenhower C, Bork P, and Segata N. *Mother-to-infant microbiome transmission from different body sites shapes the developing infant gut microbiome*. Cell Host & Microbe

- Featured on the journal cover.

2018 Yassour M, Jason E, Hogstrom L, Siljander H, **Ferretti P**, Asnicar F, Segata N, Vlamakis H, Lander E, Huttenhower C, Knip M, and Xavier R. *Strain-level identification of mother-to-child bacterial transmission during the first few months of life.* Cell Host & Microbe

2018 Dassi E*, **Ferretti P***, Covello G, Bertorelli R, Denti MA, De Sanctis V, Tett A, and Segata N. *The short-term impact of probiotic consumption on the oral cavity microbiome.* Scientific Reports

2017 Duranti S, Lugli GA, Mancabelli L, Armanini F, Turroni F, Kieran James K, **Ferretti P**, Gorfer V, Milani C, Zolfo M, Viappiani V, Pasolli E, Masè C, Postai D, Tateo S, Pedrotti A, Segata N, Van Sinderen D, and Ventura M. *Maternal inheritance of bifidobacterial communities and bifidophages in infants through vertical transmission*. Microbiome

2017 Asnicar F, Manara S, Zolfo M, Truong DT, Scholz M, Armanini F, **Ferretti P**, Gorfer V, Pedrotti A, Tett A, and Segata N. *Studying vertical microbiome transmission from mothers to infants by strain-level metagenomic profiling*. mSystems

2016 **Ferretti P**, Farina S, Cristofolini M, Girolomoni G, Tett A, and Segata N. *Experimental metagenomics and ribosomal profiling of the human skin microbiome.* Experimental Dermatology

2015 Milani C, Mancabelli L, Lugli G, Duranti S, Turroni F, Ferrario C, **Ferretti P**, Gorfer V, Tett A, Segata N, Van Sinderen D, and Ventura M. *Exploring vertical transmission of the gut microbiota from mother to child.* Applied and Environmental Microbiology

2015 Da Costa ARF, Fedrizzi T, Lopes ML, Monica Pecorari M, Wanda Lailan Oliveira da Costa, **Ferretti P**, Olivier Jousson, Nicola Segata, and Enrico Tortoli. *Characterization of 17 strains belonging to the Mycobacterium simiae complex and description of Mycobacterium paraense sp. nov.* International Journal of Systematic and Evolutionary Microbiology

2014 Tortoli E, Fedrizzi T, Pecorari M, Giacobazzi E, De Sanctis V, Bertorelli R, **Ferretti P**, Tagliazucchi S, Jousson O, and Segata N. *The new phylogenesis of the genus Mycobacterium.* International Journal of Systematic and Evolutionary Microbiology

FUNDING

Pending:

K99/R00 NIH Pathway to Independence Award 2025-2030
PI: P. Ferretti
Impact of microbial transmission and diet on preterm infant health
Total Award Amount: $971,042

Contributed:

Kenya Institute of Primate Research (KIPRE), Kenya. 2025-2026
PI: M. Chege, CO-I: P. Ferretti
Impact of diet and social structure on gut microbiome and antimicrobial resistance in captive Baboons at the Kenya Institute of Primate Research
Total Award Amount: $20,000
Role: Conceptualization, co-developed study design and analysis plan, co-wrote first draft.

Awarded:

Scholarship 2017
Department of Cellular, Computational and Integrative Biology, University of Trento, Italy.
Total Award Amount: 6,000€ (Decreto n. 259 dd 19.12.2016)
Role: Research Assistant

Scholarship 2016
Department of Information Engineering and Computer Science, University of Trento, Italy.
Total Award Amount: 6,000€ (Decreto n. 134 dd 21.04.2016)
Role: Research Assistant

PRESENTATIONS

Invited Talks

- Eco-evolutionary modeling of host-symbiont communities workshop, Banff International Research Station (BIRS), Banff, Canada, February 2026.
- Department of Biology, Loyola University, IL, USA, April 2024.
- Centre for Microbiology and Environmental Systems Science (CMESS), University of Vienna, Austria, February 2024.
- Department of Medicine, University of Chicago, IL, USA, August 2022.
- Stanford University School of Medicine, IL, USA, April 2022.
- Leiden University Medical Centre, Netherlands, April 2021.
- Quadram & Earlham Institute, University of Norwich, UK, November 2020.
- Comano Thermal Center Research, Trento, Italy, May 2016.

Contributed Talks

- Microbiome Center, Department of Medicine, University of Chicago, Illinois, USA, October 2024.
- Microbiome Virtual International Forum (MVIF) n.9, October 2022.
- The 2nd Theodor Escherich Symposium on medical microbiome research, University of Graz, Austria, May 2015.

Posters (presenting author)

- Human Milk Institute Symposium (HMI26), University of California San Diego, CA, USA, Mar 2026.
- Microbiome Center, University of Chicago, IL, USA, April 2025.
- Microbiome Conference, Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, USA, October 2024.
- The Human Microbiome Conference, EMBL, Heidelberg, Germany, September 2023.
- International Human Microbiome Congress (IHMC22), Kobe, Japan, November 2022.
- International Society for Microbial Ecology (ISME18), Lausanne, Switzerland, August 2022.
- International Human Microbiome Congress (IHMC18), Killarney, Ireland, June 2018.
- The Human Microbiome Conference, EMBL, Heidelberg, Germany, September 2018.
- Symbiomes: Systems Biology of Host-Microbiome Interactions, Pultusk, Poland, June 2015.

HONORS & AWARDS

- Poster Award at the Human Milk Institute Symposium (HMI26), San Diego, USA, Mar 2026.
- ISME Scholar Mobility Fund, Nov 2025.
- Travel Award for The Human Microbiome Conference, Heidelberg, Germany, August 2023.
- Poster Award at International Society for Microbial Ecology (ISME18), Lausanne, Switzerland, August 2022.
- Poster Award at The Human Microbiome Conference, EMBL, Heidelberg, Germany, September 2018.
- Merit Award for the Master of Science in Cellular and Molecular Biotechnology, June 2017
- EMBO Travel Grant for Symbiomes: Systems Biology of Host-Microbiome Interactions, Pultusk, Poland, June 2015.

WORKSHOPS & TRAINING

- Machine Learning and AI Hackathon (3 days), University of Chicago, IL, USA, June 2025.
- Machine Learning and AI Workshop (2 weeks), University of Chicago, IL, USA, June 2025.

- Theory of Microbial Symbiosis Workshop (3 weeks), University of Hawai'i, USA, Feb 2024.
- Deep Learning and Neural Networks Workshop, EMBL Heidelberg, Germany, November 2022.

TEACHING EXPERIENCE

Guest Lecturer: "Microbiome Transmission in Social Groups"
Department of Biology, Loyola University, IL, USA, 2024.

Teaching Assistant: Data visualization
European Molecular Biology Laboratory (EMBL) Heidelberg, Germany, 2022.

Guest Lecturer: "Microbiome in infant's health and clinical recommendations"
University of Trento, Italy, 2021.

Teaching Assistant: Python programming
European Molecular Biology Laboratory (EMBL) Heidelberg, Germany, 2018.

Guest Lecturer: "The Human Microbiome in Thermal Medicine"
"II level master in Hydrology and Thermal Medicine", Comano Terme, Trento, Italy, April 2017.

Guest Lecturer: "The Human Microbiome"
"II level master in Hydrology and Thermal Medicine", Comano Terme, Trento, Italy, April 2016.

Private Tutor
Master's and Bachelor's students in Computer Science and Management Engineering, including students with blindness, low vision, dyscalculia, and dyslexia, 2014-2016.

Guest Lecturer and Teaching Assistant: "Shotgun Metagenomics and Genome Assembly Approaches"
WebValley Summer School, University of Trento and Fondazione Bruno Kessler, Trento, Italy, 2014.

MENTORSHIP EXPERIENCE

Mary Chege **October 2023 – Present**

Mary is a graduate student jointly affiliated with the Akinyi Lab at the University of Nairobi, Kenya, and the Archie Lab at the University of Notre Dame, US. I have been mentoring Mary on numerous projects on wild baboons in the Amboseli ecosystem, including the profiling of their gut microbial and eukaryotic communities, and the analysis of microbiome and antimicrobial resistance transmission patterns both within baboon social groups and between baboons and livestock.

Tess Brunnen **June 2024 – Present**

Tess is a graduate student in the Blekhman Lab at the University of Chicago. I worked with Tess during her rotation in the Blekhman Lab, where she characterized the breast milk microbiome and metabolome and investigated their impact on infant gut microbiome composition. I continue to mentor her on her current project as a graduate student, focused on microbiome signatures associated with *Pseudomonas aeruginosa* infection in the hospital setting.

John James Colgan **February 2024 – Present**

JJ is a graduate student in the Blekhman Lab at the University of Chicago. His research focuses on characterizing the microbiome of non-Westernized human communities, with particular emphasis on populations affected by konzo, an irreversible paralytic disease. I mentor JJ on the reconstruction of metagenome-assembled genomes (MAGs) to identify previously uncharacterized members of the gut microbiome in these communities, as well as on his analysis of microbial strain-sharing across villages.

Mattea Allert **February 2023 – December 2024**

I mentored Mattea during her graduate studies in the Blekhman Lab at the University of Minnesota, where she investigated the composition and functional characteristics of the human breast milk microbiome. She is now a postdoctoral scholar at the University of Minnesota.

Askarbek Orakov **October 2020 – June 2022**

Askarbek completed his graduate studies in the Bork Lab at EMBL Heidelberg, where I worked with him on his meta-analysis of microbiome composition associated with the presence of *Fusobacterium* species, a clinically relevant bacterial clade. He is currently a postdoctoral scholar in the Huttenhower Lab at Harvard University.

EXTERNAL SCIENTIFIC OUTREACH & SERVICE

Public Outreach

- Penne Amiche della Scienza - Science correspondence program for students. Oct 2025 – Present
- South Science Festival, Science Slam Finalist, University of Chicago, Illinois, USA. Oct 2024
- Scienza in Gioco, Rome, Italy - outreach seminar series for high school students, virtual. May 2024

Media & Interviews

- Interview for the Italian Newspaper "Gazzetta dell'Emilia" Dec 2025
- Spotlight Article, University of Chicago. Mar 2024
- Book Interview, conducted by Roberta Raffaeta', Cultural Heritage at Cà Foscari Jun 2020
- University of Venice, Italy. Book title: "Antropologia dei microbi. Come la metagenomica sta configurando l'umano e la salute" - ISBN 9788879756983 2020.
- Regional News Coverage "Trentino Buongiorno Regione", Trento, Italy. Jun 2018

Conference Organization and Chairing

- The Microbiome Center (TMC) Symposium, University of Chicago, Illinois, USA. Apr 2025
- Microbiome Virtual International Forum (MVIF), Pacific Committee Member Feb 2023 – Present

Advisory Groups and Professional Memberships

- Applied Microbiology International (AMI) - One Health Advisory Group Member Mar 2026 – Present
- Member of The International Data Reuse Consortium. Jun 2024 – Present
  (consortium led by Prof. Alex Probst, University of Duisburg-Essen)
- The Wildlife Society (TWS). Aug 2025 – Present
  Wildlife Diseases and Molecular Ecology Working Groups
- International Society for Microbial Ecology (ISME). Aug 2023 – Present

Peer Review

- Invited to author a commentary article on a published work in *mSystems*. Sept 2025
- Invited reviewer for >30 manuscripts for 14 accredited journals: 2020 – Present
  Nature (co-review), Nature Communications, Cell Host & Microbe, Nature Biofilms and Microbiomes, mSystems, ISME Journal, Cell Reports, Critical Reviews in Food Science and Nutrition, PLOS Computational Biology, FEMS Microbiology Ecology, Current Opinion in Microbiology, PeerJ, PLOS One, and Scientific Reports.